

Mapping the Leaf: From Chá to Tea

How ancient trade routes shaped how we say “tea”

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I. INTRODUCTION & MOTIVATION

Originating from China, tea is celebrated as a popular drink across numerous global cultures. Despite the vast language diversity represented by these cultures, linguists have noticed a pattern in how we say “tea”. The majority of languages that have a word for tea can be classified into two categories: either deriving from a variation of the word “cha” or “te”. The two variations of the word tea originate from two Chinese language variations, where Min Nan languages, typically spoken along the coast, refer to tea as “te”, and the wider Sinitic family of Chinese languages, typically spoken in the mainland, use “cha”.

Interestingly, when observing the distribution of the two “tea” variates, a pattern appears. Clusters of the varieties have formed that linguistics have theorized are due to historic trade routes, such as the Silk Road and the Dutch East India Company trade routes. That is, the variety that is spoken by a certain language is dependent on where in China the language’s predominate region first received tea.

Most noticeably, the clusters of “cha” variants appear to be concentrated in Eurasia, while clusters of “te” variants appear in Europe and the Southeast Asian Austronesian islands. The clusters of variants have lead linguistics to further theorize that “cha” variants were the result of land-based trade routes, while “te” variants appeared from sea-based trade routes, coining the phrase “Tea, if by sea. Cha, if by land” [6].

Using the word variates and trade route theory as a base, this project aims to expand the understanding of how trade routes have shaped the varieties and if any additional variates may appear that may further explain the origin of tea globally. Specifically, this project uses historical milestones to orient the timeline of feasible global spread possibilities and integrate word forms into the visualization for users to be able to explore the global map and potentially uncover additional patterns that have not yet been explored by existing studies.

Coming from an undergraduate degree in linguistics, I hoped to be able to apply data visualization techniques into a topic that I’ve been interested in exploring. I’ve started a similar personal project in the past to explore the distribution of the “tea” form variants, but fell short on the strategy of how to visualize the data. Now, using my data visualization

coursework, I hope that I’m able to paint a story for the te/cha phenomenon that provides meaningful insights into the work that linguistics and historians have completed.

II. RELATED WORK

A. *World Atlas of Language Structures*

Likely the most notable work that has been conducted on the distribution of the “tea” form varieties (and the basis of this project as well as many other studies) is provided by the World Atlas of Language Structures (WALS) [7]. The WALS dataset has compiled existing research from several linguistic studies to provide a fairly robust dataset of languages and their classification into either deriving from the Sinitic form “cha”, the Min Nan form “te” or falling into an “Other” category if the word form doesn’t appear to inherently derive from either variations. The dataset also includes geographical data that was used in creating a map visual for the varieties with color coding to denote the variation. A photo of the map is included below for reference.

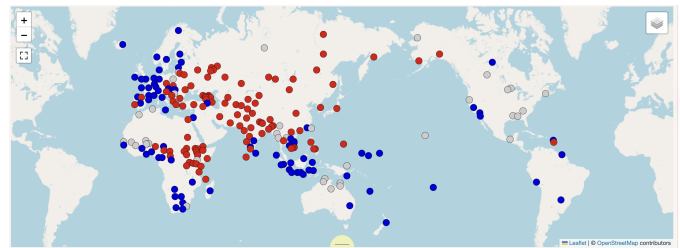


Fig. 1. WALS Tea Derivations Map

B. *Expansions of WALS*

While searching for additional data to create a more robust version of the WALS dataset visualization, I came across two articles [6] [4], which used the WALS data but added additional features. Both visualizations added a network component to demonstrate how the varieties had spread, but Lokhov’s [4] article stood out to me in that it had some word forms integrated into the visualization as tooltips.

These two visualizations played into my inspiration for further expanding the existing work completed, while also giving me additional data to include that was available to the public for downloading.

C. Lexibank and Historical Context

As an addition to the WALS dataset, Lexibank [3] is a similar database for linguistic data that compiles various studies to create word form datasets for various concepts. One of the concepts that Lexibank has available is "Tea", which lists languages and their word form for "tea", as well as providing geospatial coordinates for mapping languages that may not be included in the WALS dataset. The visualization provided by the Lexibank database focuses on displaying language family categorizations, and it does not explicitly state whether the word forms are derived from "cha" or "tea". However, the addition of this dataset allowed for a more diverse set of data points to expand the WALS data coverage.

For historical context, Griffiths' [2] book provides an overview of the history of tea, explaining from how tea evolved from a local medicinal drink to a lucrative global trade good. A majority of the historical milestones were borrowed from Griffiths' work.

Finally, in order to visualize the journey of tea as a trade good and how trade influenced the word forms, I searched for trade data that could be integrated into the map. Using the historical milestones as a basis for the timeline and major events that played into tea's distribution, I found that the Silk Road and the Dutch East India Trading Company were likely the most influential trade routes. Appleton's [5] visualization on ArcGIS that has been integrated into National Geographic's Silk Road interactive map provided geospatial data for mapping the routes. However, the Dutch East India Trading Company routes did not appear to have geospatial readily available that could be integrated into a map. Instead, I was able to use the ports listed on Ahn's [1] map image in combination with Python's searoute package to recreate the routes in an interactive map.

D. Adaptions from Related Work

While the WALS dataset and related articles were the basis for this project, the existing works were all missing one part of the puzzle that I hoped to address. By combining the related works into one visual, my goal is to paint a clear picture of how actual trade routes and historic data were the drivers of how different languages say "tea" today. In addition to the historical context, another gap I aimed to address was the missing word form data in the WALS dataset. While the word forms aren't critical in understanding the distribution of the derivations, I thought it would add another layer of interest to users and potentially lead to insights about the "Other" category or additional patterns that have evolved.

One such pattern that was able to be addressed with including the word forms is the appearance of the "chai" variation. Although derived from "cha", "chai" is a widespread variant throughout the Caucus region and Southwest Asia that could lead to further insights about the trade routes and the global spread of tea.

III. DATA DESCRIPTION

A. Datasets Used

The predominant datasets used were a combination of the WALS and Lexibank datasets with the addition of Lokhov's dataset, which is adapted from the WALS set and includes a few word forms that were excluded in the Lexibank set. Additionally, Appleton's geospatial dataset for the Silk Road route was used to recreate the trade routes.

B. Preprocessing

In order to preprocess the data to be used in the visualization, the first step was to retrieve and navigate through Lexibank's broader dataset to isolate word forms for just the "tea" concept. After isolating the tea entries, the WALS dataset was also read as a CSV file. Unnecessary columns were then dropped before joining the two datasets using the language name as the joining key. In order to keep consistency, the latitude and longitude coordinate columns were collated, using the WALS coordinates as the precedence. The same collation procedure was then performed on the language name and language family columns to create a consistent and consolidated merged dataset. Additionally, there were a few languages in the Lexibank dataset that had interchangeable words for "tea", which were removed from the dataset to avoid confusing the visualization.

Afterwards, the dataset was saved as a csv, and a few operations were performed in sheets to insert word forms from Lokhov's dataset that weren't covered by Lexibank. I also located any instances where Lexibank and WALS language names may have had inconsistencies and combined them into the same entry. Then, any additional missing word forms that were easily searchable and verifiable by creditable sources were added and word forms that weren't initially included in the WALS dataset were classified. Word forms that appeared in one category frequently were used as a basis for filling in missing categorizations with similar word forms, as well as the language family classification. That is, missing derivation categorizations were filled using a combination of similar word forms' categorization and categorizations of languages belonging to the same family. Any word forms or categorizations that could not be reasonably deduced were omitted from the visualization. Finally, the word forms were added into a separate column, along with their Romanized spelling if the languages are written in a script for the visualization's tooltip.

As an additional step for the expansion of existing work, words derived from "cha" were also subcategorized into the "chai" derivation classification according to their word forms. After processing, the WALS dataset was successfully able to be increased from 230 entries to 276.

C. Challenges

While integrating the WALS and Lexibank datasets were integral in being able to add the context of word forms

without having to manually search and add them in entry-by-entry, their combination did pose a few challenges. The main challenge that became apparent during processing was the lack of overlap for quite a few data points. This hurdle was mostly able to be alleviated by using a combination of Lakhov's dataset and the reasonable deductions mentioned above. However, there were several data points that did not have word forms readily available that were listed as a variation of their underlying derivation rather than a specific word form for the tooltip. Similarly, several Lexibank data points ended up being omitted due to their word forms missing derivation classifications and not following any closely related patterns that could be used to deduce their classification. Despite the complexity of joining the two datasets, 46 additional languages were able to be combined with the original WALS dataset to create a more robust, diversified dataset.

IV. DESIGN & METHODOLOGY

In order to construct the visualization, I implemented an interactive global map utilizing the Folium library in Python. Because of the scale of the data and its categorical clustering, a map seemed as though it would be the best option for clearly visualizing the "networks" of word form derivations. Additionally, using a map as a base would allow users to be able to orient themselves with where languages are based regionally without having to overcrowd the visual with additional labels. The hope was to be able to maximize what users would be able to deduce from the visualization, while minimizing the amount of "ink" or additional information on the screen that would distract from the goal of the visual and potentially cause a cognitive overload.

Keeping the reduction of cognitive load in mind, the main visual encoding tool that was utilized was color and shape to distinguish between the word form categories. I settled on using a variation of a diverging color palette, where "te" variations were colored green-blue, "cha" and "chai" variations were colored red-orange and the other category fell between the spectrum at a light-yellow. Because I still wanted to be able to add a secondary distinction between the "cha" and "chai" variations, I decided that the strongest way to visualize that the two still belong to the same overarching origin, while distinguishing between the two was to keep a similar shade so that they would visually belong together. So, to signal to users that the "chai" category is just a variation of "cha", I decided to use a secondary category encoding method, using shape to denote "chai" as a square, while "cha" and the other two branches of tea derivations are all marked with circles.

Now that the datapoints were plotted and encoded on the map, the trade routes also needed to be visualized. Thinking about how I could best paint the story of how trade affected the evolution of tea word forms, I thought of using distinct but

similar shades to the existing categorical shades. Specifically, since a majority of the words that derive from "te" were influenced by the Dutch East India Trading Company, I thought it would strengthen the visualization to plot this trade route using a darker shade of the blue-green that was used for the "te" points. Similarly, the Silk Road trade routes were drawn using a darker red-orange. I also thought this would help ease the strain of encoding several categories into the map as to not introduce too many color variations that may become confusing to discern.

With trying to reduce cognitive load in mind, because the datapoints and trade routes can be overwhelming to the user if the user would like to hone in on one specific aspect of the map, each of the derivation categories, the trade routes and milestone markers were added as individual feature groups in Folium. Introducing the feature groups allows users to interact with the map's layers by selecting or deselecting which groups to display. For example, if a user wanted to hide the "Other" category, they could toggle the visibility of these datapoints by selecting or deselecting the category from the layer control panel. Using HTML unicode and hexcolor codes, the layer control panel was also able to act a legend for the map by displaying the category icons and their color adjacent to the category name. An example of the mechanism is displayed in the figure below, where the "chai" and "other" categories are toggled to be hidden.

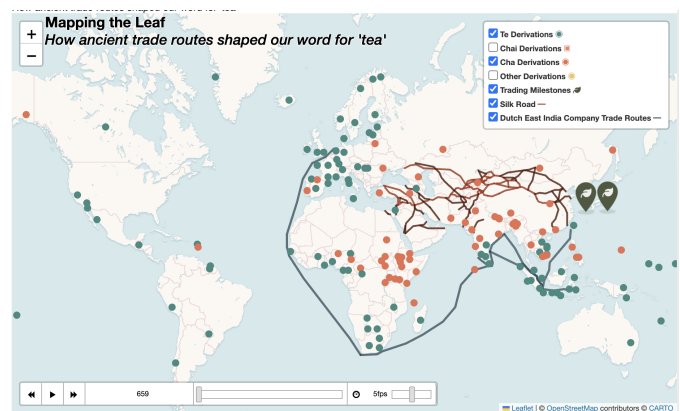


Fig. 2. Map Legend Toggle

In addition to the trade routes, one of the goals for this visualization was to be able to add in historic milestones to orient the context of the data. Using Folium's marker feature, I first added a few milestones that I was able to gather from my research. However, adding the markers with direct labeling on top of all of the existing data points would become too crowded. At the same time, without the date displayed or label of the marker, I realized the milestones wouldn't achieve the goal of providing the additional historic context, unless a user went through every milestone's tooltip individually, which could cause an issue in accessibility and cognitive load in trying to remember what each milestone had noted. Thinking about how I could compromise to remedy

the main issues of both of these approaches, I thought adding animations to introduce the timeline aspect would achieve a middleground in that the context would be apparent visually without having too much additional direct labeling that could overwhelm the visual. Therefore, I added an animation mechanism so that the milestone markers would appear chronologically with the approximate year displayed in the animation toggle. Explanations for the specific events are still held in the tool tip, though the consideration here was that each milestone marks a point where tea is introduced to a region, so even without the tooltips' explanations, the main idea would still be delivered. The milestones were also added as a layer that could be hidden or expanded in case the milestone markers were to block any datapoints of interest.

Additional considerations made were to contain the languages and their word forms and language families in tool tips for a similar reason as the milestone markers. Avoiding the direct labeling would drastically improve the visual clarity and would not take away from the main idea of the visualization given the categorical encoding set in place via the color and shape channels. I also decided to overlay the data over a blank map to further decrease the visual crowding that may occur with city names and streets that are often embedded into Folium's base map selections. The last consideration I implemented was including the title of the map across the top left of the map to provide context for those viewing the visualization for the first time, and I enabled the legend to be draggable in case while interacting with map, the legend blocks accessibility to viewing certain portions of the map.

Included below is an example of the draggable legend and milestone tooltip display.

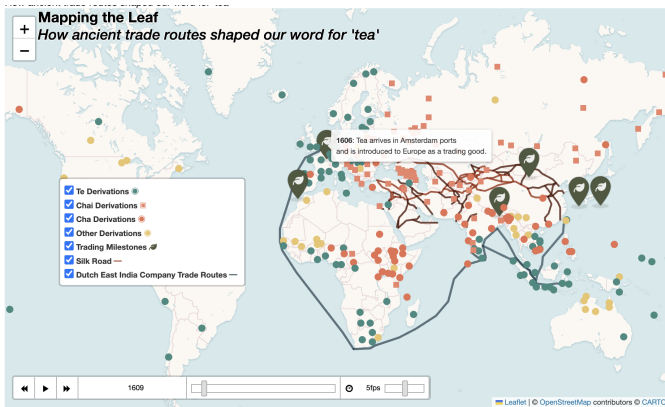


Fig. 3. Legend Movement and Milestone Tooltip

V. IMPLEMENTATION

Once the data was prepared and the design choices were considered, the visualization was ready to be implemented in Python. After initial research on platforms that are used for building interactive maps, I decided on using the Folium

library in Python with a Google Colab notebook. Plotting the datapoints onto the map was a fairly simple process with the guide of Folium's documentation. First, the prepared dataset was split into four mini-datasets, one for each of the derivation categories. Then, feature groups were created for each category so that they may be toggled individually and the datapoints were iterated through to create markers for the map before being added to their constituent feature group.

Once the datapoints were plotted, drawing the trade routes was a slight hurdle since Folium's polyline function creates straight lines from one coordinate to the next and most trade routes aren't direct shots from one hub to another. After considering options to be able to display the routes, I settled on using Appleton's [5] geospatial dataset for the Silk Road on ArcGIS, which I could request directly into my notebook and pass to Folium's polyline function to draw the routes. However, once I came to the step of plotting the Dutch East India Trading Company routes, I ran into an issue of there not being a geospatial equivalent readily available like the Silk Road. The alternative I discovered was the searoute library on Python that inputs coordinates of a port of origin and a port of arrival and outputs the resulting vector of feasible sea routes. Using a reference map [1] for the Dutch East India Trade Company maritime routes, I was able to search the coordinates for the major ports along the route and use searoutes to plot the path.

As the last step to integrate the historical milestones, I utilized the TimestampedGeoJson package from Folium's plugins library to map the milestones to their constituent regional coordinates and the approximate year they occurred. Conveniently, TimestampedGeoJson automatically handles the animation aspect of the plot.

VI. RESULTS & INSIGHTS

A. Probable Routes

As a result of the research combined with the visualization, I was able to deduce that the Silk Road and Dutch East India Company trade routes were likely the most influential in deciding which variety for "tea" a language may use. Although the British East India Company had similar routes to the Dutch around the same time period, the Dutch trade routes aligned the most with the distribution of the varieties, as well as the timeline of the milestones.

B. Origin of Chai Variety

While I wasn't able to definitively decide on the origin point of the "chai" variety branch, the visualization seems to suggest that the variant occurred as tea was being traded between Northern China and the Eurasian groups that lived near the Altai Mountain region. Many of the early civilizations that established early trade relationships with China lived around this area, where there appears to be a divide of "cha" derivation towards the East and "chai" variations towards the West.

C. Additional Pattern Discoveries

Another interesting phenomenon that I was able to discover with the inclusion of word forms is that several Southeast Asian languages fall into the "Other" category but seem to have a pattern in common. Rather than having a "ch" or "t" sound at the beginning of their word for tea, these languages have adopted an "l" sound. For example, there are a few languages that use "la" or "lu" at the start of their word for tea. This may suggest that these languages may not have an equivalent of a "ch" sound and adopted a close approximation of a sound, perhaps uncovering an additional linguistic pattern, though further phonetic studies would need to be conducted. A few examples of the "la"/"lu" pattern are included below.



Fig. 4. "la"/"lu" Pattern (SE Asia)

Similarly, I also discovered a cluster of African languages that fall into the "Other" category, who add a vowel before a variation of a "ti" or "tay" sound, such as "i" or "a". This could potentially result in additional classifications of several African languages as deriving from "te" if phonetic studies could be conducted to confirm that the overall word form is still a variation of "te".

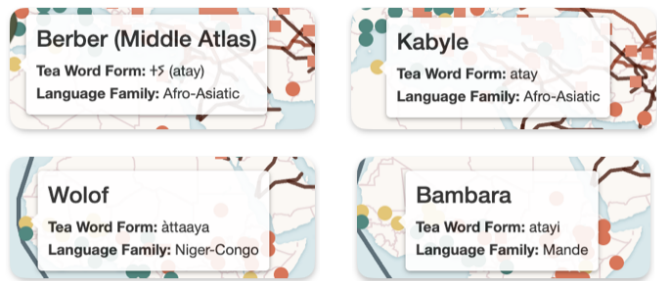


Fig. 5. "A(ti/tay)" Pattern (NW Africa)

As a final thought, the visualization with trade routes provides a fairly confident view that most languages in Eurasia and South Asia were introduced to tea from the Silk Road, as well as several African countries on the West coast that are in close proximity to the Middle East. For Europe, the Southeast Asian Austronesian islands and the Eastern coast Africa, these cultures most likely were introduced to tea from the Dutch East India Trading Company.

VII. LIMITATIONS & FUTURE WORK

One limitation that became apparent when conducting research prior to beginning the visualization is the lack of collated data in one single source. While previous research has been conducted on each of the aspects included in the visualization individually, there does not appear to be a single source that combines word forms, derivations and dates when tea may be estimated to be adopted into a language. Similarly, since tea culture began in China in the early centuries, there is much uncertainty in the exact timeline of events with the lack of documentation from ancient times. While existing historic timelines were able to give a broad picture of the timeline of the spread of tea, I was unable to find a definitive source for exact timelines when tea was introduced to various countries globally. Having milestones for various regions definitely helps to paint a picture of the evolution of tea, but if there were a dataset for estimated introductions of tea into a country, the visualization could become even stronger to draw an even clearer picture of the spread of tea and where exactly different languages received their word for tea.

Drawing off of this idea of having exact timelines, future expansions to this project could include a comprehensive study of tea trade to reconstruct a fairly comprehensive timeline that could be turned into a dataset for this purpose. With additional time, this would be an interesting adaption to analyze if any other patterns appear, such as uncovering how languages that fall into the "Other" category may have adopted a different derivation, or if certain locations were hubs for transforming the word form by adding an additional letter for example. A definitive timeline could theoretically be used to reconstruct a network visualization displaying estimated distribution routes or hubs over time.

Additionally, further linguistic studies could be conducted to examine the phonetic features of the languages mentioned in the result section that shared underlying patterns but fell into the "Other" category. Existing phonetic studies could be analyzed to determine if Southeast Asian and African languages may adhere to phonological rules that dictate if the initial sounds may need to be preceded by a vowel or if they're restricted from "ch" or "t" sounds. Further studies into these variations could either reveal a new category of variations or lead to additional subcategorizations into the "cha" or "te" derivations.

VIII. CONCLUSION

In conclusion, this visualization project aimed to further existing studies, namely the WALS dataset, by introducing additional layers of linguistic and historical context. Though theories exist as to how trade routes have shaped how we say "tea" globally, there is a lack of collated studies that combine both linguistic and historical features into one bundle.

Incorporating historical milestones and literature research, I was able narrow down two most likely trade routes that

were influential drivers in the linguistic distribution of "tea". Regions that fell along the Silk Road (Eurasia, South Asia and Western Africa) tend to demonstrate the "cha" variety, suggesting they may have shared the same origin for tea in Northern China. Conversely, locations that hosted or were near neighbors of Dutch East India Trading Company ports share the "te" word derivation, namely Europe, the South Eastern Austronesian islands and Eastern Africa.

Additionally, with the added layer of linguistic word forms, there appear to be three subcategories- the known "chai" variation and two additional patterns that have yet to be explored. One of the clusters is gathered in Northwest Africa with the potential of being categorized as a "te" variant with additional phonetic study. The other cluster is located in Southeast Asia with an unknown origin, though following a similar pattern of starting with "la" or "lu".

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